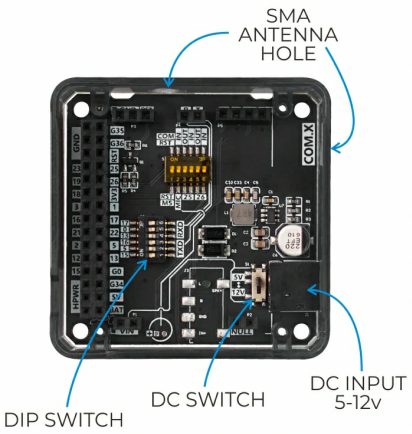
Module COMX Cat1

SKU:M031-H



















Description

Module COMX Cat1 is an LTE CAT1 data communication module, equipped with the SIM-A7680C module that supports LTE CAT1 frequency bands in China, and supports LTE-TDD/LTE-FDD wireless communication standards. The module integrates an SMA external antenna interface to ensure communication quality and signal stability. It also includes a DC 5 ~ 12V power input interface, providing more

power supply options. Compatible with the M5Stack stacking system, it can be stacked with other modules in the series for functional expansion. Compared to NB-IoT, CAT1 offers stronger data transmission capabilities, with a maximum downlink rate of 10Mbps (uplink) / 5Mbps (downlink). This module is ideal for applications requiring data transmission rates, such as remote control and data passthrough.

Features

- SIM-A7680C
  - Supports CAT1 frequency bands in China
  - LTE CAT1 module, supports LTE-TDD/LTE-FDD wireless communication standards
  - Maximum downlink rate of 10Mbps / maximum uplink rate of 5Mbps
  - UART interface / AT command control
  - Nano SIM card slot
- Module interfaces:
  - 1 x DC 5/12V POWER INPUT
  - SMA antenna interface
  - Base integrated with a DIP switch for communication pin switching
- 5 x 5cm size, compatible with M5Stack stacking system
- Development platforms:
  - UIFlow
  - Arduino

Includes

- 1 x Module COMX Cat1
- 1 x L-shaped hex key 1.5mm (compatible with M2 screws)
- 1 x SMA antenna

Applications

- Remote control / data collection
- Cloud data applications

Certifications

SIM-A7680C Module Certification

- 3C/SRRC/NAL/RoHS/REACH

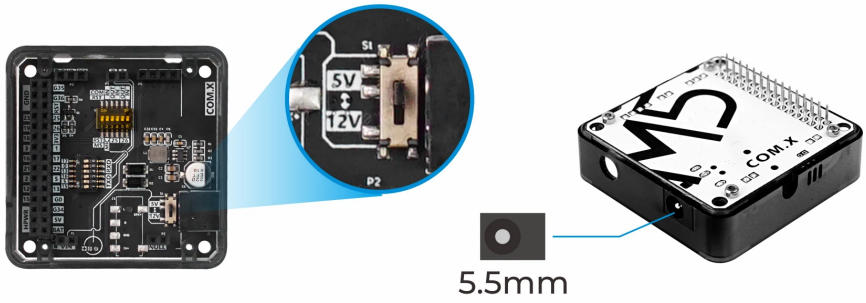
Specifications

Specification	Parameters
Communication Module	SIM-A7680C
Uplink/Downlink Speed	Maximum downlink rate 10Mbps / Maximum uplink rate 5Mbps
Supported CAT1 Bands	LTE-TDD: B34/B38/B39/B40/B41 LTE-FDD: B1/B3/B5/B8
Network Protocols	TCP/IP/IPV4/IPV6/MultiPDP/FTP/HTTP/DNS/RNDIS/ECM/PPP/TLS/LBS/TTS/MQTT/WiFi Scan
Communication Method	UART 115200bps 8N1
SIM Card Slot	Nano
Power Input	M5-Bus(5V) / DC INPUT(5V/12V)
Product Size	54.2 x 54.2 x 13.2mm
Net Weight	28.6g
Package Size	165 x 60 x 36mm
Gross Weight	78.0g

Learn

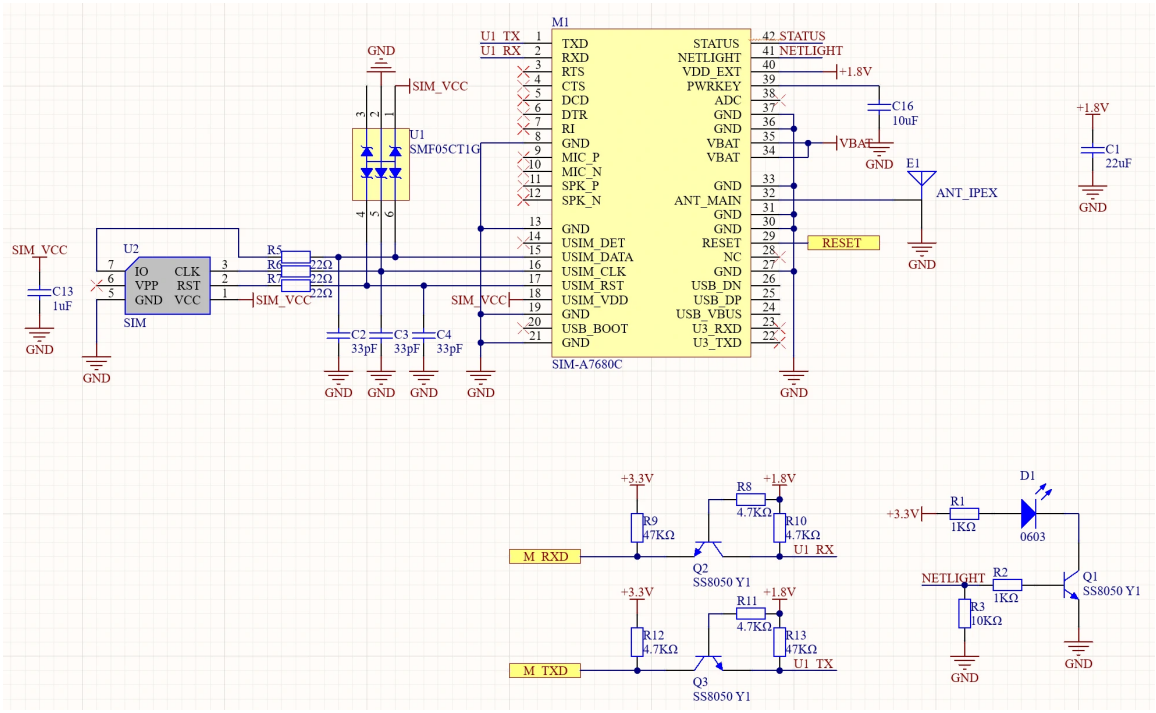
Power Supply Switching

The module base has a DC power input interface. When using this interface to connect power, strictly adhere to the input range (5-12V) to prevent module damage. The internal power DIP switch can adjust the voltage level of the internal VIN terminal to adapt to different modules.



Schematics

Module COMX Baseboard Schematic



PinMap

M5-Bus

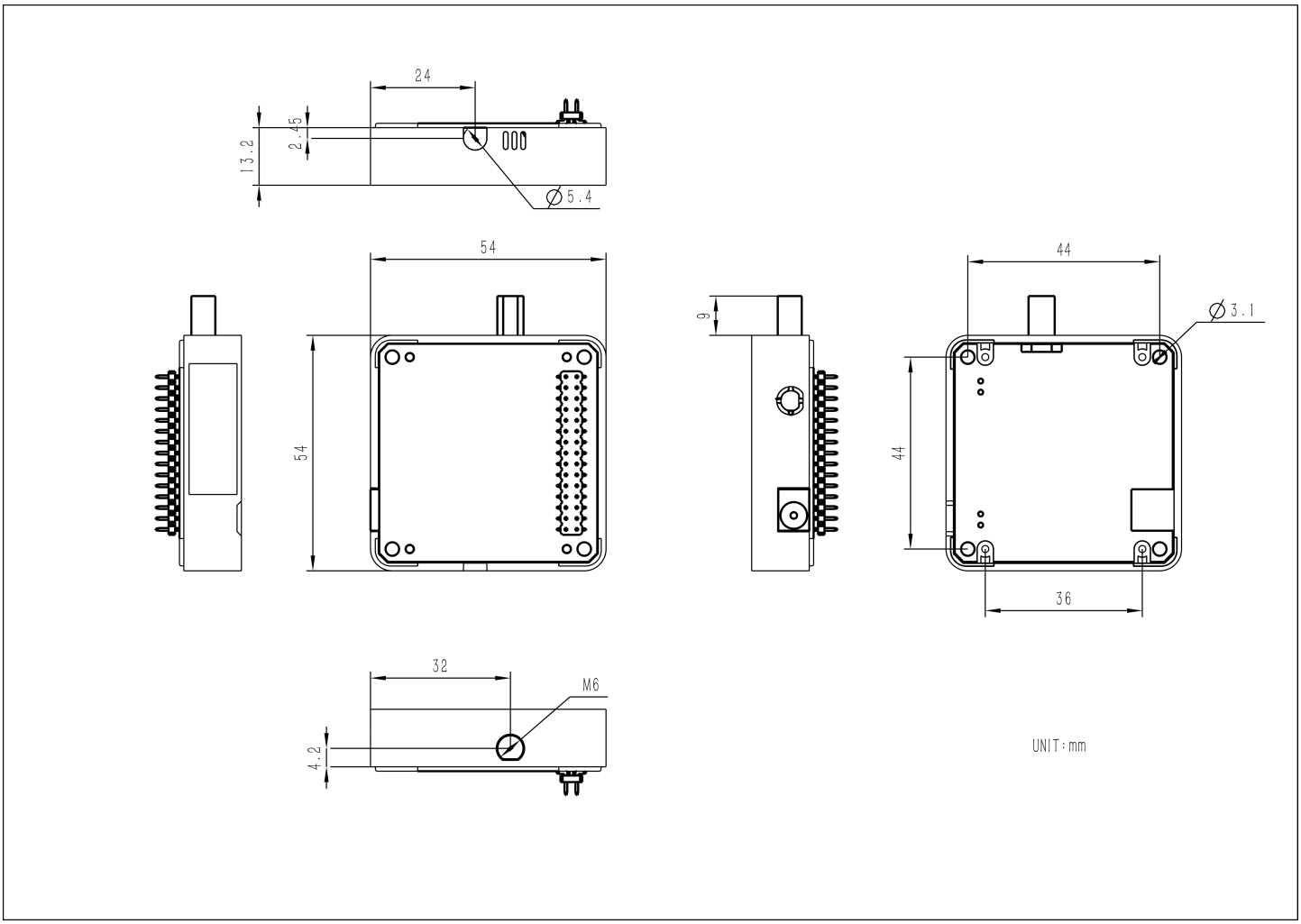
Switch

The pins marked `SW` on the M5-Bus below can be switched via DIP switches to adapt to different main controllers.

PIN	LEFT	RIGHT	PIN
GND	1	2	
GND	3	4	
GND	5	6	
	7	8	
	9	10	
	11	12	3V3
	13	14	
TXD (SW)	15	16	RXD (SW)
	17	18	
	19	20	TXD (SW)
	21	22	RXD (SW)
TXD (SW)	23	24	RXD (SW)
	25	26	
	27	28	5V
	29	30	

Model Size

- Module COMX Cat1 Model Size PDF



Datasheets

- Application
  - A76XX Series_Audio_Application Note_V1.02.pdf
  - A76XX Series_Blue Tooth_Application Note_V1.02.pdf
  - A76XX Series_CTBURST_Application Note_V1.00.pdf
  - A76XX Series_CTBURST_Application Note_V1.00.pdf
  - A76XX Series_FOTA_Application Note_V1.00.pdf
  - A76XX Series_FTP(S)_Application Note_V1.02.pdf
  - A76XX Series_GNSS_Application Note_V1.02.pdf
  - A76XX Series_GNSS_Dynamic_Loading_Instructions_Application Note_V1.00.pdf
  - A76XX Series_HTTP(S)_Application Note_V1.02.pdf
  - A76XX Series_LBS_Application Note_V1.02.pdf
  - A76XX Series_MQTT(S)_Application Note_V1.02.pdf
  - A76XX Series_SSL_Application Note_V1.02.pdf
  - A76XX Series_Sleep Mode_Application Note_V1.02.pdf
  - A76XX Series_TCPIP_Application Note_V1.02.pdf
  - A76XX Series_UART_Application Note_V1.02.pdf
  - A76XX Series_UIM HOT SWAP_Application Note_V1.02.pdf
  - A76XX Series Cloud Platform Protocol Application Document_V1.01.pdf
- AT Command

- A76XX Series MQTT_EX_AT Command Manual_V1.00.pdf
- A76XX Series_AT_Command_Manual_V1.06.pdf

Softwares

Arduino

- Examples
  - Module COMX Cat1 MQTT Client with M5Core
  - Module COMX Cat1 MQTT Client with M5Core2
- Libraries
  - TinyGSM
  - PubSubClient

UiFlow1

- Module COMX Cat1 UiFlow1 Docs